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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 1: Matrices

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - definitions, types, operations, determinant, inverse, and solving simultaneous equations, with worked examples and exam practice.

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> **How to use these notes:** Read one section at a time, copy each worked example into your notebook, then try the practice questions on the last pages without looking at the answers. Matrices is the first and one of the most scoring units in the Class 9 paper - most questions are short and follow fixed steps.

What this unit covers

1. What a matrix is - rows, columns, order and elements
2. Types of matrices - row, column, square, null, diagonal, scalar, identity, symmetric, skew-symmetric and transpose
3. Equality, addition, subtraction and scalar multiplication
4. Multiplication of matrices and its laws
5. Determinant, singular and non-singular matrices, adjoint
6. Inverse of a 2 x 2 matrix
7. Solving simultaneous linear equations - inverse method and Cramer's Rule
8. Formula summary, common mistakes, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 1 (book pages 1-49). The notes follow the book's order and syllabus. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. What Is a Matrix?

A **matrix** is a rectangular arrangement of numbers (or symbols) in rows and columns, enclosed in brackets. Matrices are used to organise data and to solve equations in a neat, short form.

Example: the marks of 2 students in 3 subjects can be written as one matrix:

Example 1. $A = \begin{vmatrix} 80 & 75 & 90 \\ 60 & 88 & 70 \end{vmatrix}$ Each **row** is one student; each **column** is one subject.

Rows, columns and order

- A **row** runs left to right (horizontal). A **column** runs top to bottom (vertical).
- The **order** of a matrix = number of rows x number of columns, read as "rows by columns".
- A matrix with *m* rows and *n* columns is called an ***m* x *n* matrix**. The example above is a 2 x 3 matrix.
- Matrices are named with capital letters: A, B, C. Their numbers are called **elements** (or entries).

Position of an element

The element in row *i* and column *j* is written $\mathbf{a_{ij}}$. The first small number is always the row, the second is always the column - remember: **row first, column second**.

Example 2. In $B = \begin{vmatrix} 4 & -1 & 7 \\ 0 & 5 & 2 \end{vmatrix}$

$b_{11} = 4$, $b_{12} = -1$, $b_{13} = 7$, $b_{21} = 0$, $b_{22} = 5$, $b_{23} = 2$. The order of B is 2 x 3, so B has 6 elements.

> **Exam trap:** Order is rows x columns, never columns x rows. A 3 x 2 matrix and a 2 x 3 matrix are different, even if they use the same numbers.

Quick check

- How many elements are in a 3 x 4 matrix? **Answer: 3 x 4 = 12.**
- In a 1 x 5 matrix, how many rows are there? **Answer: 1.**

2. Types of Matrices (Part 1)

Type	Meaning	Example
Row matrix	Only 1 row, any number of columns. Order 1 x n.	[3 -2 5 0]
Column matrix	Only 1 column, any number of rows. Order m x 1.	Column with entries 2, -4, 6 (3 x 1)
Rectangular matrix	Rows and columns are not equal (m is not equal to n).	A 2 x 3 or a 4 x 2 matrix
Square matrix	Rows = columns. Order n x n.	A 2 x 2 or a 3 x 3 matrix
Null / Zero matrix	Every element is 0. Written O.	$\begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$
Diagonal matrix	A square matrix in which every element outside the main diagonal is 0.	$\begin{vmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{vmatrix}$

The main diagonal

In a square matrix, the diagonal from the top-left corner to the bottom-right corner is the **main (principal) diagonal**. Its elements are a_{11}, a_{22}, a_{33}, and so on - the elements where the row number equals the column number.

> **Remember:** Only square matrices have a main diagonal, a determinant, and - when the conditions are met - an inverse. Row, column, rectangular and null matrices cannot be square unless their order happens to be n x n (a null matrix can be square or rectangular).

Worked check

Classify $\begin{vmatrix} 9 & 0 \\ 0 & 9 \end{vmatrix}$: it is square (2 x 2), diagonal (zeros off the diagonal), and - as the next page shows - also a scalar matrix and a multiple of the identity matrix.

3. Types of Matrices (Part 2)

Type	Meaning	Example / Note
Scalar matrix	A diagonal matrix whose diagonal elements are all equal.	Diagonal entries 4, 4, 4; all other entries 0
Identity / Unit matrix	A scalar matrix whose diagonal elements are all 1. Written I.	$I_2 = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$
Symmetric matrix	A square matrix equal to its own transpose: $A = A^t$. Here $a_{ij} = a_{ji}$.	$\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}$
Skew-symmetric matrix	A square matrix with $A^t = -A$. Its diagonal elements are always 0.	$\begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix}$

Transpose of a matrix

The **transpose** of A, written $\mathbf{A^t}$ (or A'), is made by turning rows into columns: the first row becomes the first column, the second row becomes the second column, and so on. The order changes from m x n to n x m.

Example 3. If $A = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix}$ (2 x 3), then $A^t = \begin{vmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{vmatrix}$ (3 x 2). Also, $(A^t)^t = A$.

Equality of two matrices

Two matrices are **equal** if and only if (1) they have the same order, and (2) their corresponding elements are equal: $a_{ij} = b_{ij}$ for every position.

Example 4. If $\begin{vmatrix} x & 3 \\ 1 & y \end{vmatrix} = \begin{vmatrix} 5 & 3 \\ 1 & -2 \end{vmatrix}$, then comparing matching positions gives **x = 5** and **y = -2**. This is a very common 2-mark exam question.

4. Addition, Subtraction and Scalar Multiplication

Addition and subtraction - same order only

Two matrices can be added or subtracted **only if they have the same order**. Add (or subtract) the corresponding elements, position by position. The result has the same order.

Example 5. Let $A = \begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix}$ and $B = \begin{vmatrix} 1 & 5 \\ -2 & 0 \end{vmatrix}$

$A + B = \begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix}$ $A - B = \begin{vmatrix} 1 & -6 \\ 5 & 4 \end{vmatrix}$

- **Commutative law:** A + B = B + A.
- **Associative law:** (A + B) + C = A + (B + C).
- **Additive identity:** A + O = A. **Additive inverse:** A + (-A) = O.
- Subtraction is **not** commutative: A - B is not equal to B - A in general.

Scalar multiplication

Multiplying a matrix by a number (a **scalar** k) means multiplying **every** element by k.

Example 6. If A is as above, then $3A = \begin{vmatrix} 6 & -3 \\ 9 & 12 \end{vmatrix}$ and $-A = \begin{vmatrix} -2 & 1 \\ -3 & -4 \end{vmatrix}$. Combined work such as 2A + 3B is done element by element in the same way.

> **Common mistake:** Trying to add a 2 x 2 matrix to a 2 x 3 matrix. Different order means the sum does not exist - in the exam, write exactly that: "Addition is not possible because the orders are different."

5. Multiplication of Matrices

Matrix multiplication is **not** element-by-element. It is **row by column**: each element of the product is the sum of (row element x column element) pairs.

When is multiplication possible?

AB is possible only when **the number of columns of A equals the number of rows of B**. If A is m x n and B is n x p, then AB is m x p. The two inner numbers must match; the two outer numbers give the order of the answer.

Example 7. $A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$, $B = \begin{vmatrix} 5 & 6 \\ 7 & 8 \end{vmatrix}$

AB, element (1,1): 1x5 + 2x7 = 19. (1,2): 1x6 + 2x8 = 22.
(2,1): 3x5 + 4x7 = 43. (2,2): 3x6 + 4x8 = 50.

So $AB = \begin{vmatrix} 19 & 22 \\ 43 & 50 \end{vmatrix}$ but $BA = \begin{vmatrix} 23 & 34 \\ 31 & 46 \end{vmatrix}$. The two products are different.

Laws of multiplication - learn the differences

- **Not commutative:** AB is generally **not** equal to BA. This is the most tested law in this unit.
- **Associative:** (AB)C = A(BC), when all products are possible.
- **Distributive:** A(B + C) = AB + AC.
- **Identity:** AI = IA = A. The identity matrix works like the number 1 in ordinary multiplication.
- **Transpose of a product:** $(AB)^t = B^tA^t$ - note the order reverses.

> **Method tip:** For each element, say it as "row times column, add the pairs". Write the pair sums first, then add - most multiplication errors come from trying to do both steps in your head.

6. Determinant, Singular Matrices and Adjoint

Every **square** matrix has a single number called its **determinant**. Class 9 works with 2 x 2 determinants.

For $A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ **det A = |A| = ad - bc** (product of the main diagonal minus product of the other diagonal).

Example 8. For $A = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix}$, |A| = (3 x 4) - (2 x 1) = 12 - 2 = **10**.

Singular and non-singular matrices

- If **|A| = 0**, A is a **singular** matrix. It has **no inverse**.
- If **|A| is not 0**, A is a **non-singular** matrix. Its inverse exists.

Quick test: $\begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix}$ has determinant (2x2) - (4x1) = 0, so it is singular.

Adjoint of a 2 x 2 matrix

For a 2 x 2 matrix there is a short rule for the **adjoint** (written adj A): **swap the main-diagonal elements, and change the signs of the other two elements**.

Example 9. If $A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then adj A = $\begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$. For the matrix in Example 8, adj A = $\begin{vmatrix} 4 & -2 \\ -1 & 3 \end{vmatrix}$.

Useful check: A x (adj A) = |A| x I. If your adjoint is correct, this product gives the determinant on the diagonal and zeros elsewhere.

7. Inverse of a 2 x 2 Matrix

The **inverse** of A, written $\mathbf{A^{-1}}$, is the matrix that gives the identity when multiplied by A: $\mathbf{A\,A^{-1} = A^{-1}\,A = I}$. It exists only when |A| is not 0.

Formula: $\mathbf{A^{-1} = (1 / |A|) \times adj\ A}$ - find the determinant, write the adjoint, then divide every element of the adjoint by the determinant.

Worked example - follow the four steps

Example 10. Find the inverse of $A = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix}$

1. **Determinant:** |A| = (3 x 4) - (2 x 1) = 10. Not zero, so the inverse exists.
2. **Adjoint:** adj A = $\begin{vmatrix} 4 & -2 \\ -1 & 3 \end{vmatrix}$ (swap 3 and 4; change the signs of 2 and 1).
3. **Divide:** $A^{-1} = (1/10) \begin{vmatrix} 4 & -2 \\ -1 & 3 \end{vmatrix} = \begin{vmatrix} 0.4 & -0.2 \\ -0.1 & 0.3 \end{vmatrix}$
4. **Verify:** $A\,A^{-1}$ should give $I = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$. Checking (1,1): 3x0.4 + 2x(-0.1) = 1.2 - 0.2 = 1. Correct.

- $(A^{-1})^{-1} = A$. $(AB)^{-1} = B^{-1}A^{-1}$ (order reverses, as with transposes).
- The inverse of the identity matrix is itself: $I^{-1} = I$.
- Always verify in the exam if time allows - verification catches sign errors in the adjoint, which are the most common mistake.

> **Never** write A^{-1} by simply flipping each element or by writing 1/a, 1/b, 1/c, 1/d. The inverse comes only from the formula above.

8. Solving Simultaneous Equations

Two equations in x and y, such as 2x + y = 5 and x + 3y = 10, can be written as one matrix equation **AX = B**, where A holds the coefficients, X holds the unknowns, and B holds the constants. Class 9 uses two methods.

Method 1 - Inverse method: $X = A^{-1} B$

Example 11. Solve 2x + y = 5, x + 3y = 10.

$A = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix}$, $|A| = 6 - 1 = 5$, adj $A = \begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix}$, so $A^{-1} = (1/5)$ adj A.

$X = A^{-1}B = (1/5) \begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} 5 \\ 10 \end{vmatrix} = (1/5) \begin{vmatrix} 5 \\ 15 \end{vmatrix} = \begin{vmatrix} 1 \\ 3 \end{vmatrix}$. **x = 1, y = 3.** Check: 2(1)+3 = 5 and 1+3(3) = 10. Both correct.

Method 2 - Cramer's Rule

For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$: **$x = D_x / D$, $y = D_y / D$**, where D is the determinant of the coefficients, D_x replaces the x-column with the constants, and D_y replaces the y-column with the constants.

Example 12 (same equations). D = 5. D_x = (5x3) - (1x10) = 5, so x = 5/5 = **1**. D_y = (2x10) - (5x1) = 15, so y = 15/5 = **3**. Same answer, as it must be.

> **If D = 0:** neither method works in this simple form - the equations have no single unique solution. In Class 9, state this clearly instead of dividing by zero.

9. Formula Summary and Common Mistakes

Idea	Rule
Order	Rows x columns. Elements: a_{ij} = row i, column j.
Addition / subtraction	Same order only; add or subtract matching positions.
Scalar multiplication	Multiply every element by the scalar.
Matrix multiplication	Columns of first = rows of second. Row x column, add the pairs. AB is not generally BA.
Transpose	Rows become columns. $(AB)^t = B^tA^t$.
Determinant (2 x 2)	$\|A\| = ad - bc$.
Singular	$\|A\| = 0$ - no inverse exists.
Adjoint (2 x 2)	Swap a and d; change the signs of b and c.
Inverse	$A^{-1} = (1/\|A\|) \times \text{adj } A$, only if $\|A\|$ is not 0.
Solving equations	$X = A^{-1}B$, or Cramer's Rule: $x = D_x/D$, $y = D_y/D$.

Five mistakes that lose marks

1. Writing the order as columns x rows.
2. Adding matrices of different orders, or multiplying element-by-element.
3. Assuming AB = BA without checking.
4. Forgetting the sign change on b and c in the adjoint.
5. Trying to find the inverse of a singular matrix ($|A| = 0$).

> **One-line revision:** Same order to add, inner numbers match to multiply, ad - bc for the determinant, swap and sign-change for the adjoint, divide by the determinant for the inverse.

10. Practice - MCQs and Questions

Multiple-choice questions

1. The order of a matrix with 3 rows and 2 columns is: (a) 2 x 3 (b) 3 x 2 (c) 6 (d) 5
2. A matrix with only one column is called: (a) row matrix (b) square matrix (c) column matrix (d) null matrix
3. If |A| = 0, then A is: (a) identity (b) singular (c) symmetric (d) diagonal
4. The transpose of a 2 x 3 matrix has order: (a) 2 x 3 (b) 3 x 2 (c) 2 x 2 (d) 3 x 3
5. For A = [4 1; 2 3] (rows: 4,1 and 2,3), |A| equals: (a) 10 (b) 14 (c) 5 (d) 12
6. AB = BA is: (a) always true (b) never true (c) not generally true (d) true for all square matrices

Short and long questions

1. Find x and y if [x 2; 3 y] = [5 2; 3 -4] (rows in brackets).
2. For A = [1 -2; 0 3] and B = [4 1; -1 2], find A + B, A - B, and 2A + B.
3. For the A and B in Question 2, find AB and BA, and state whether AB = BA.
4. Find the determinant and the inverse (if it exists) of C = [2 1; 5 3].
5. Solve by the inverse method or Cramer's Rule: 3x + y = 7, x + 2y = 9.

Answer key

MCQs: 1(b), 2(c), 3(b), 4(b), 5(a) - |A| = 12 - 2 = 10, 6(c).

Questions: 1) x = 5, y = -4. 2) A+B = [5 -1; -1 5], A-B = [-3 -3; 1 1], 2A+B = [6 -3; -1 8]. 3) AB = [6 -3; -3 6], BA = [4 -5; -1 8] - not equal. 4) |C| = 6 - 5 = 1, C^{-1} = [3 -1; -5 2]. 5) x = 1, y = 4 (check: 3+4 = 7, 1+8 = 9).

> **Final tip:** In the board paper, show every step - order, determinant, adjoint, then the answer. Step marks are given even when a final number slips, but only if the steps are written.